

Indian Institute of Technology Jodhpur

Course Name: Time Series Analysis Assignment-1

Select a time series from “Quandl” package. Then copy its short URL and import the data using
`y <- Quandl(“????”, api_key= “????”, type= “ts”)` (Replace each “????” with the appropriate values.)

1. Plot graphs of the data.
2. Create appropriate train and test sets.
3. Try to identify an appropriate ARIMA model based on the train set.
4. Do residual diagnostic checking of your ARIMA model. Are the residuals white noise?
5. Use your chosen ARIMA model to forecast for the length of test set.
6. Now try to identify an appropriate ETS model based on the same train set.
7. Do residual diagnostic checking of your ETS model. Are the residuals white noise?
8. Use your chosen ETS model to forecast for the length of test set.
9. Which of the two models do you prefer and why?
10. Do you change your conclusion if the cross validation error is used as a measure of accuracy?